

**DEFENDING CYBORGIA:
A COMPREHENSIVE GUIDE TO MITRE ATT&CK
MAPPING**

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I. INTRODUCTION

The Kingdom of Cyborgia and the Role of the Castellan: The Kingdom of Cyborgia is under attack. As the newly appointed Castellan, the primary responsibility is to secure the Royal Citadel against the encroaching Shadow Guild. This guide serves as a pedagogical bridge between the game's narrative elements and the real-world **MITRE ATT&CK** (Adversarial Tactics, Techniques, and Common Knowledge) framework.

Objective of the Game: The core objective is to strategically allocate the King's Gold to construct defenses that mitigate specific cyber-threats. Players must operate within a limited budget each round, making critical decisions about which vulnerabilities to prioritize. While a chosen defense remains a permanent fixture of the castle once built, the Shadow Guild consistently evolves its tactics, employing increasingly devious methods to breach the perimeter. Success in Cyborgia is measured by the efficacy of the defensive layer; a round is considered successful when the player has implemented at least 70% of the correct defenses required to mitigate the active threat.

II. HIGH-LEVEL MAPPING: NARRATIVE TO TECHNICAL

The following matrix provides a quick-reference guide for educators to understand how narrative attacks and defenses align with the MITRE ATT&CK framework.

Narrative Story	MITRE ID	Real-World Equivalent	Narrative Defenses
The Poisoned Messenger	T1566	Phishing	Village Drills, Kennel Hounds, Scribe's Ledger, Portcullis Toll
The Undercover Spy	T1552	Unsecured Credentials	Stone Wall Reinforcement, Scribe's Ledger, Royal Guard, Iron-Bound Chests, Village Drills
The Corrupt Seneschal	T1098	Account Manipulation	The Two-Key Vault, Stone Wall Reinforcement, Royal Guard, Iron-Bound Chests
The Trojan Crate	T1027	Obfuscated Files	The Kennel Hounds, Scribe's Ledger, Village Drills
The Sorcerer's Script	T1059	Command and Scripting	The Royal Guard, Kennel Hounds, Scribe's Ledger, Portcullis Toll

The Bewitched Relic	T1204	User Execution	The Suspicion Ward, The King's Decree, The Blacksmith's Seal, The Tower Ballistas, The Portcullis Toll, The Village Drills
The False Raven	T1557	Adversary-in-the-Middle	The Village Drills, The Walled Districts, The Tower Ballistas, The Privy Council, The Customs Officers, The Ciphared Scrolls, The Sealed Catacombs
The Merchant's Disguise	T1071	Application Layer Protocol	The Customs Officers, The Tower Ballistas
The Peasant Stampede	T1499	Endpoint Denial of Service	The Customs Officers
The Smuggler's Tunnel	T1048	Exfiltration Over Alternative Protocol	The Steward's Roster, Iron-Bound Chests, The Walled Districts, The Tower Ballistas, The Customs Officers

III. IN-DEPTH THREAT PROFILES: THE SHADOW GUILD'S ARSENAL

1. The Poisoned Messenger (Phishing: T1566)

- **Narrative:** A courier delivers a letter that appears official, but the ink is infused with a toxin designed to incapacitate the very first individual who handles the parchment.
- **Technical Context:** This mimics **Phishing**, where attackers send fraudulent communications that appear to come from a reputable source. The goal is to steal sensitive data or install malware on the victim's machine.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1566/>

2. The Undercover Spy (Unsecured Credentials: T1552)

- **Narrative:** A member of the Shadow Guild infiltrates the castle in the guise of a servant, searching for physical keys left on hooks or eavesdropping on whispered secrets in the corridors.

- **Technical Context:** This represents the search for **Unsecured Credentials**. Attackers often look for passwords stored in plain text files, environment variables, or insecurely handled tokens within a network.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1552/>

3. The Corrupt Seneschal (Account Manipulation: T1098)

- **Narrative:** An enemy agent attempts to bribe the castle's record-keeper to alter the laws of the realm. Success allows the agent to grant themselves "Duke" status, providing them with unrestricted access to the castle gates.
- **Technical Context:** This mirrors **Account Manipulation**, specifically privilege escalation. Attackers modify account permissions to maintain access or gain higher-level administrative rights.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1098/>

4. The Trojan Crate (Obfuscated Files: T1027)

- **Narrative:** A merchant enters the castle with a crate of supposedly standard supplies. In reality, the crate houses a magical parasite engineered to spread through and weaken the castle's stone structure.
- **Technical Context:** This mimics **Obfuscated Files or Information**, where malicious code is hidden within seemingly benign files or through encryption to evade detection by security software.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1027/>

5. The Sorcerer's Script (Command and Scripting Interpreter: T1059)

- **Narrative:** An enemy mage stands beyond the castle walls, chanting ancient, complex commands. These incantations are designed to exploit the mechanical logic of the drawbridge, forcing it to lower against the will of the defenders.
- **Technical Context:** This represents the use of a **Command and Scripting Interpreter**. Attackers use scripts (like PowerShell or Python) to execute commands that bypass security controls and automate their attacks.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1059/>

6. The Bewitched Relic (User Execution: T1204)

- **Narrative:** The Guild leaves a beautifully crafted, seemingly harmless music box in the bustling courtyard. Overcome by curiosity, a young stablehand turns the crank, unwittingly triggering a dormant spell that unbars the barracks doors.
- **Technical Context:** Adversaries may rely on a user to execute a malicious file or run commands for initial access or lateral movement. This typically happens when a user

opens an email attachment, runs a macro in a document, or double-clicks a disguised executable file.

- **MITRE Link:** <https://attack.mitre.org/techniques/T1204/>

7. The False Raven (Adversary-in-the-Middle: T1557)

- **Narrative:** Shadow Guild rogues quietly intercept messenger birds flying between the kingdom's watchtowers. They meticulously swap the real scrolls with forged orders and send the birds on their way, manipulating the guard patrol routes from the shadows.
Technical Context: Adversaries may position themselves between two or more networked devices to intercept, monitor, alter, or redirect communications. This allows them to harvest sensitive credentials, session tokens, or manipulate payloads in transit.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1557/>

8. The Merchant's Disguise (Application Layer Protocol: T1071)

- **Narrative:** Shadow Guild spies masquerade as regular apple merchants and bards, passing secret hand signals and hidden messages right under the noses of the guards in the bustling town square.
Technical Context: Adversaries may communicate using application layer protocols to blend their command and control (C2) traffic with normal, legitimate network activity. Common protocols abused include HTTP, HTTPS, DNS, and SMTP.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1071/>

9. The Peasant Stampede (Endpoint Denial of Service: T1499)

- **Narrative:** The Guild pays a massive mob of fake peasants to crowd the drawbridge and riot at the gates, completely freezing castle operations and preventing legitimate royal messengers from entering or leaving.
- **Technical Context:** Adversaries may conduct Denial of Service (DoS) attacks targeted at endpoints to degrade, disrupt, or completely exhaust the availability of services, applications, or system resources, rendering them unavailable to legitimate users.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1499/>

10. The Smuggler's Tunnel (Exfiltration Over Alternative Protocol: T1048)

- **Narrative:** Instead of trying to carry stolen gold out through the heavily guarded main gate, thieves bypass the checkpoints by sneaking the kingdom's secrets out through sewer grates and underground rivers.
- **Technical Context:** Adversaries may steal or exfiltrate data from a victim network using a different protocol than the one originally used to compromise or control the system. This helps them bypass standard data loss prevention (DLP) filters.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1048/>

IV. IN-DEPTH DEFENSE PROFILES: THE ROYAL ARSENAL

1. The Two-Key Vault (Multi-factor Authentication: M1032)

- **Mechanism:** This defense requires both a physical wax seal and a secret whisper to gain entry. Even if a guard's physical keys are stolen by an intruder, the vault remains securely locked.
- **Cybersecurity Equivalent: Multi-factor Authentication (MFA).** It requires users to provide two or more verification factors to gain access to a resource.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1032/>

2. The Royal Guard (Privileged Account Management: M1026)

- **Mechanism:** These are elite soldiers assigned specifically to shadow high-ranking castle officials. Their sole purpose is to ensure that only the most trusted individuals in the kingdom handle the Royal Scepter.
- **Cybersecurity Equivalent: Privileged Account Management (PAM).** This involves monitoring and protecting accounts that have elevated permissions to prevent unauthorized use.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1026/>

3. The Portcullis Toll (Web Restrict: M1021)

- **Mechanism:** A checkpoint established at the very edge of the kingdom's borders. It is designed to stop merchants arriving from known "enemy lands" and to forbid the entry of strange or unidentified crates.
- **Cybersecurity Equivalent: Web Restrict.** This includes web filtering and firewalls that block access to malicious domains or prevent the download of unauthorized content.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1021/>

4. Iron-Bound Chests (Permissions: M1022)

- **Mechanism:** Sensitive scrolls and documents are locked within specific rooms of the keep. Under this system, a cook has no need for a key to the armory, and a stablehand is denied access to the Royal Treasury.
- **Cybersecurity Equivalent: File and Directory Permissions.** This follows the principle of "least privilege," ensuring users only have access to the data necessary for their specific roles.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1022/>

5. The Kennel Hounds (Antivirus/Antimalware: M1049)

- **Mechanism:** These beasts are trained to detect the scent of poisoned items, hidden daggers, or any "unnatural" objects brought into the citadel.
- **Cybersecurity Equivalent: Antivirus (AV).** AV software scans for known signatures of malicious software and monitors for suspicious behavior on a system.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1049/>

6. Stone Wall Reinforcement (OS Hardening: M1028)

- **Mechanism:** Specialized laborers fill in ancient cracks in the masonry, seal off unused drainage pipes, and remove hidden "thief holes" that have existed for generations.
- **Cybersecurity Equivalent: Operating System (OS) Hardening.** This process involves securing a system by reducing its surface of vulnerability, such as by disabling unused services or closing unnecessary ports.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1028/>

7. The Scribe's Ledger (Audit: M1047)

- **Mechanism:** A dedicated monk stationed at the gates meticulously records every individual who enters, leaves, or moves between the different rooms of the castle. While this ledger cannot physically stop a crime, it provides a perfect record of exactly who was involved after the event occurs.
- **Cybersecurity Equivalent: Audit/Logging.** Detailed logs allow security teams to track the "who, what, and when" of a system breach for forensic analysis.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1047/>

8. Village Drills (User Training: M1017)

- **Mechanism:** Regular town hall meetings are held where the peasantry is educated on common ruses. They are taught that the King will never use a random messenger bird to request their gold coins.
- **Cybersecurity Equivalent: User Training.** Security awareness training helps employees recognize phishing attempts and social engineering tactics.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1017/>

9. The Scribe's Vault (Data Backup: M1053)

- **Mechanism:** Keeps reserve copies of the kingdom's records so losses can be restored after an attack.
- **Cybersecurity Equivalent: Data Backup.** Creates restorable copies of critical data.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1053/>

10. The Outland Post (Remote Data Storage: M1029)

- **Mechanism:** Stores copies far from the castle so a local disaster cannot destroy everything at once.
- **Cybersecurity Equivalent: Remote Data Storage.** Off-site data storage to prevent local loss.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1029/>

11. The Royal Inquisitor (SSL/TLS Inspection: M1020)

- **Mechanism:** Examines sealed deliveries for hidden dangers before allowing them into the keep.
- **Cybersecurity Equivalent: SSL/TLS Inspection.** Decrypts traffic to find hidden threats.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1020/>

12. The Royal Lineage (Active Directory Configuration: M1015)

- **Mechanism:** A strict hierarchy of nobles and titles is enforced. A mere squire cannot simply declare themselves a Duke to command the castle guards or access the treasury.
- **Cybersecurity Equivalent: Active Directory Configuration.** Configures directory services securely to prevent privilege escalation and credential theft.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1015/>

13. The Steward's Roster (User Account Management: M1018)

- **Mechanism:** The Steward actively revokes castle access from mercenaries whose contracts have expired and banishes idle loiterers from the courtyard.
- **Cybersecurity Equivalent: User Account Management.** Manages the lifecycle of user accounts to limit unnecessary or stale privileges.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1018/>

14. The Riddler's Gate (Password Policies: M1027)

- **Mechanism:** Guards demand a complex, ever-changing passphrase from travelers—no more using "password" or the King's birthday to gain entry.
- **Cybersecurity Equivalent: Password Policies.** Enforces strong password complexity, length, and rotation requirements.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1027/>

15. The Walled Districts (Network Segmentation: M1030)

- **Mechanism:** The castle is divided into isolated rings. Even if the outer bailey falls to the Shadow Guild, the inner keep remains sealed behind raised drawbridges.
- **Cybersecurity Equivalent: Network Segmentation.** Divides a network into smaller, isolated segments to contain lateral movement during a breach.

- **MITRE Link:** <https://attack.mitre.org/mitigations/M1030/>

16. The Tower Ballistas (Network Intrusion Prevention: M1031)

- **Mechanism:** Vigilant archers actively scan the horizon, immediately shooting down any incoming projectiles, siege weapons, or suspicious ravens before they land.
- **Cybersecurity Equivalent: Network Intrusion Prevention.** Monitors network traffic to actively detect and block malicious exploits.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1031/>

17. The Blacksmith's Seal (Limit Software Installation: M1033)

- **Mechanism:** Only tools and weapons forged by the official royal armory are allowed. Peasants are forbidden from bringing their own makeshift swords into the barracks.
- **Cybersecurity Equivalent: Limit Software Installation.** Restricts users from installing unauthorized software applications.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1033/>

18. The Privy Council (Limit Access to Resource Over Network: M1035)

- **Mechanism:** Only designated, high-ranking advisors are allowed into the map room; the doors are completely hidden and walled off from the rest of the kingdom.
Cybersecurity Equivalent: Limit Access to Resource Over Network. Restricts access to specific network resources based on logical business needs.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1035/>

19. The Customs Officers (Filter Network Traffic: M1037)

- **Mechanism:** Guards rigorously inspect all incoming merchant carts and outgoing messengers, turning away anyone lacking the proper royal travel permits.
- **Cybersecurity Equivalent: Filter Network Traffic.** Filters incoming and outgoing network traffic based on IP addresses, ports, or protocols.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1037/>

20. The King's Decree (Execution Prevention: M1038)

- **Mechanism:** A magical ward that instantly paralyzes anyone attempting to cast unauthorized spells or read from forbidden tomes within the castle walls.
- **Cybersecurity Equivalent: Execution Prevention.** Blocks unauthorized scripts, binaries, or untrusted code from executing on endpoints.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1038/>

21. The Suspicion Ward (Behavior Prevention on Endpoint: M1040)

- **Mechanism:** Guards closely monitor the behavior of the staff. If the court jester suddenly starts drawing a sword instead of juggling, he is tackled immediately.

- **Cybersecurity Equivalent: Behavior Prevention on Endpoint.** Monitors and blocks active processes exhibiting suspicious or anomalous behavior patterns.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1040/>

22. The Ciphered Scrolls (Encrypt Sensitive Information: M1041)

- **Mechanism:** Royal decrees are written in a dead, magical language. Even if a Shadow Guild spy steals the scroll, they cannot read a single word of it.
- **Cybersecurity Equivalent: Encrypt Sensitive Information.** Encrypts sensitive data at rest or in transit to protect confidentiality from unauthorized eyes.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1041/>

23. The Sealed Catacombs (Disable or Remove Feature or Program: M1042)

- **Mechanism:** Laborers purposely collapse old, forgotten tunnels and board up abandoned secret passages so thieves cannot use them to bypass the gates.
- **Cybersecurity Equivalent: Disable or Remove Feature or Program.** Disables or removes unnecessary software features or protocols to reduce the overall attack surface.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1042/>

24. The Royal Wax Seal (Code Signing: M1045)

- **Mechanism:** Every official missive must bear the King's enchanted wax seal. Forgeries lacking the proper crest are immediately thrown into the hearth.
- **Cybersecurity Equivalent: Code Signing.** Digitally signs software to verify the identity of the author and ensure it hasn't been altered.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1045/>

25. The Mason's Upkeep (Update Software: M1051)

- **Mechanism:** Stonemasons constantly patch weathering walls, reinforce rusting iron gates, and upgrade the defenses against newly discovered Shadow Guild siege tactics.
- **Cybersecurity Equivalent: Update Software.** Applies security patches to software and operating systems to remediate known vulnerabilities.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1051/>

26. The Keep's Protocols (Software Configuration: M1054)

- **Mechanism:** Strict rules govern the castle's daily operations. Fires must be extinguished at night, and weapons must be locked away, ensuring the castle operates safely.
- **Cybersecurity Equivalent: Software Configuration.** Configures software applications securely to minimize corporate risk and enforce safety baselines.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1054/>

V. GAME DOWNLOAD AND DISTRIBUTION

For Mac Users

1. Download and unzip the MacOS package.
2. Run the following command in the Terminal:

Shell

```
xattr -dr com.apple.quarantine MITREGame.app
```

3. Launch the game from the folder.

For Windows Users

1. Download and unzip the Windows OS package.
2. Run MITRE Game.exe to begin.

What is included in the downloaded file(s):

```
├─ MacOS/
│   ├── MITRE Game/           # Supporting assets for MacOS
│   ├── MITRE Game           # MacOS Application executable
│   ├── MITRE Game Teacher Resource Guide.pdf
│   └─ MITRE Game User Guide.pdf
├─ Windows OS/
│   ├── MITRE Game.exe       # Windows Application executable
│   ├── MITRE Game Teacher Resource Guide.pdf
│   └─ MITRE Game User Guide.pdf
├─ README.md                 # Project overview and setup
└─ MITRE Game User Manual.pdf # Full system documentation
```

VI. IN-DEPTH GAMEPLAY GUIDE

We recommend that educators guide students through the following gameplay loop:

- **Step 1: Preparation.** Students should first familiarize themselves with the narrative-to-technical mappings found in this guide to understand the relationship between the Shadow Guild's actions and the MITRE ATT&CK framework.
- **Step 2: Initiation.** To begin the simulation, click the **Play** button on the main menu.
- **Step 3: The Building Phase & Intelligence Gathering.** Each level starts in the Building Phase. Players are presented with an **Intelligence Report** that provides a narrative description of the incoming threat.

- **Step 4: Strategic Defense Selection.** While staying within the allocated **King's Gold** budget for the round, players must choose the defenses they deem most effective for protecting the Royal Citadel.
- **Step 5: Lock Defenses.** Once defenses are placed on the game board, the player must **lock** them into position. Once built, these defenses become a permanent fixture of the castle's perimeter.
- **Step 6: The Attack Phase & Outcomes.** The game transitions to the Attack Phase to test the efficacy of the player's defensive layer.
 - **Level Failure:** If the player fails to implement at least 75% of the correct defenses required to mitigate the active threat, they lose the level. In this case, they are provided with an increased budget for a second attempt and a hint regarding a specific effective defense.
 - **Level Victory:** If the player successfully repels the attack, they progress to the next level and repeat the cycle starting from the Building Phase (Steps 3–5).

VII. CONCLUSION

Defending Cyborgia is an iterative process. Students must understand that no single defense is a silver bullet; rather, a "defense-in-depth" strategy—combining technical hardening (Stone Walls) with user awareness (Village Drills) and vigilant monitoring (The Scribe's Ledger)—is required to repel the Shadow Guild successfully. By mapping these metaphors to the MITRE ATT&CK framework, students gain a foundational understanding of modern cybersecurity resilience.